LISTEN:
A Framework for Multi-Objective Selection

From language to personalized choices

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Which trade-off does this traveler prefer?

PREFERENCE UTTERANCE

"Maximize time in NYC; arrive early enough to sleep. Prefer a direct flight."

→ **LLM decision assistant**

AIRLINE	ARRIVAL	STOPS	PRICE
United	6:52 p.m.	Nonstop	$386
Delta	**7:45 p.m.**	**Nonstop**	**$372**
JetBlue	9:08 p.m.	Nonstop	$218
American	6:40 p.m.	1 stop	$294

RECOMMENDED

Delta · $372

Nonstop
Arrives 7:45 p.m.

People already use LLMs to make choices

25%

of surveyed travelers use generative AI for travel research.

Deloitte · 2026 Summer Travel Survey

42%

of surveyed consumers have used AI assistants for online shopping.

Adobe Digital Insights · August 2026

How well do these tools match one person's preferences?

[1] Deloitte, 2026 Summer Travel Survey, p. 5.

[2] Adobe Digital Insights, AI Traffic Trends Report, August 2026, pp. 5, 7.

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Three prompt strategies

Full batch

Choose from the full set of options.

The model organizes the selection.

LISTEN-T
(Tournament)

Compare small batches.

Multi-round tournament.

LISTEN-U
(Utility)

Infer and refine utility weights.

scores the full set of options and adjust scoring

When does orchestration help

Full Batch: Simplest method

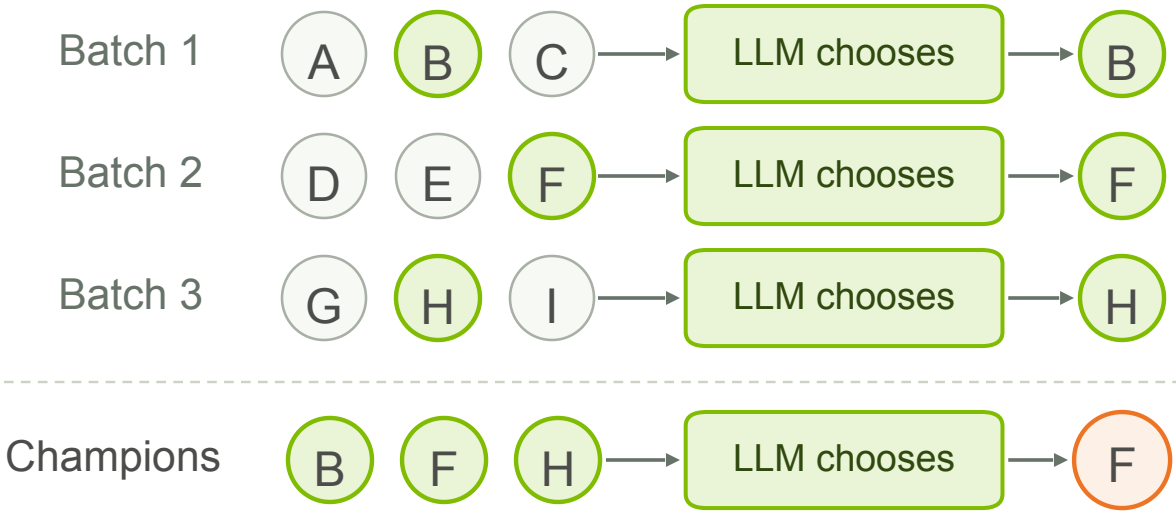


- Give the LLM the prompt and all available options.
- Let it compare trade-offs using the stated preferences.
- Return its selected option in a single call.

LISTEN-T: tournament selection

LLM sees: utterance + one small random batch

LLM outputs: one winner



Final winner

LISTEN-U: refines a utility function

LLM sees
utterance +
feature definitions

→ LLM reasons →

LLM outputs
one weight per
numerical attribute

Scoring equation: score(item) = ∑(weight × features)

Set of options

A [$294, 4h, 1]
B [$218, 2h, 0]
C [$403, 2h, 0]

→ **Code scores** all items →

Scores



→ Current best Item B

Next LLM call sees: utterance + current weights + current best

Outputs: revised weights → code rescores ↺

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LISTEN study: 3 preference scenarios

Survey data

Flights: Selecting the best flight based on preferences

107 preference utterances

Collected from prolific

Custom data

Exam Scheduling: 1 preference utterance. Thousands of options and uncommon/less data task

Headphones: 1 preference utterance designed to challenge the LLM with many specific constraints

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Survey Flights: preferences and rankings

100 prompts + their writers' top-20 rankings

1. **Describe a trip**
   Write a preference utterance.
2. **Collect flights**
   Retrieve matching flight options.
3. **Rank top 20**
   The writer supplies ground truth.

EXAMPLE PROMPT · ABBREVIATED

DC → Seoul, July 6. Cheapest fare + one checked bag.

DCA/IAD. At DCA, avoid rush hour 2–3 h before takeoff.

≤1 stop; ≤20 h if connecting. Pay ≤$150 extra for nonstop.

US airline or Korean Air; pay ≤$100 extra for Korean Air.

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Human baseline: rank for someone else

“Human rerank” = a person predicts the prompt writer’s choices.

1 **Read someone else’s prompt**
Show the rater the preference prompt written by the original traveler.

2 **Review the same flight options**
Show the flight options that were available to the prompt writer.

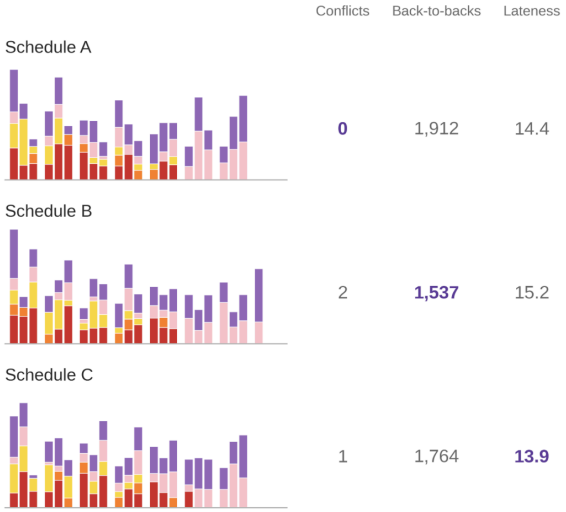
3 **Rank on the prompt writer’s behalf**
Predict the writer’s preferences—not the rater’s own.

HUMAN BASELINE
Evaluate the rater’s choices against the writer’s own ranking.

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Headphones + Exam Scheduling

- **Exam scheduling:** Cornell Spring 2025 data; rankings by Adam.
- **Headphones:** Amazon options; preferences and rankings by Francis.
- For both, 5 of us read the prompts and tried to rank on their behalf.



Equal weights: a no-LLM baseline

Can we beat simple heuristics?

1 STANDARDIZE

$z = (x - \mu) / \sigma$

μ = feature mean; σ = standard deviation over the full set of options.

2 ALIGN DIRECTION

Minimize? Use −z.

Keep z if higher is better; negate z if lower is better.

3 WEIGHT EQUALLY

Score = average

Average the aligned scores. Pick the highest average.

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Evaluation metric: normalized average rank (NAR)

NAR = average of [rank of selected item ÷ N]

N = total number of options; rank 1 is the writer's favorite.

Only the top m are ranked. An unranked choice gets the mean remaining rank: (m + 1 + N) / 2.

Lower is better.

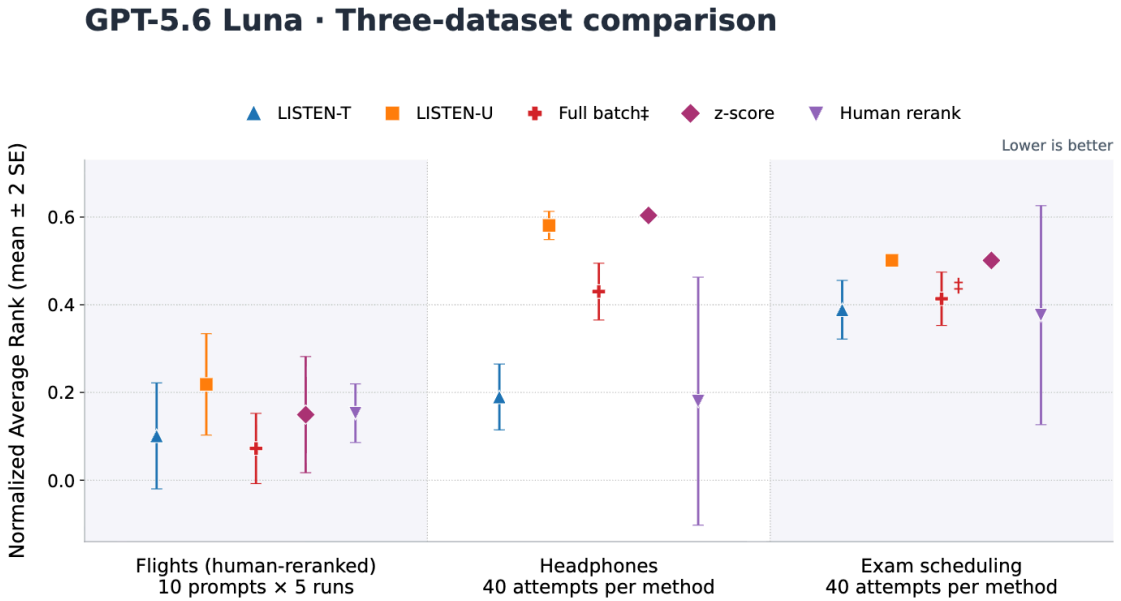
Example: rank 5 out of 100 → 0.05

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Results: LISTEN improves selection

LISTEN-T and full batch are better than LISTEN-U and z-score

LISTEN-T does best on Headphones because non-linear preference for brand, color etc.



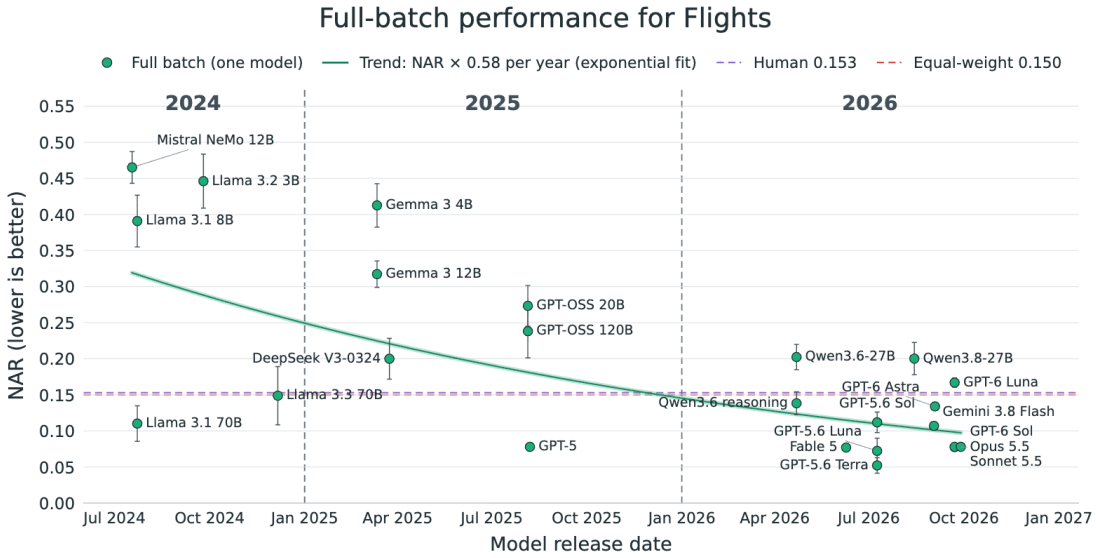
Original LISTEN evaluation. Equal weights = z-score average. Means and ±2 SE intervals unchanged.

LLMs have become super-human for flights

This year simply asking an LLM to pick a flight for you was better than the humans in our survey.

For our Flights in particular full batch has significantly improved.

1. Larger context window
2. Reasoning



All 24 model configurations · 10 fixed prompts · error bars ±1 SE across seed blocks

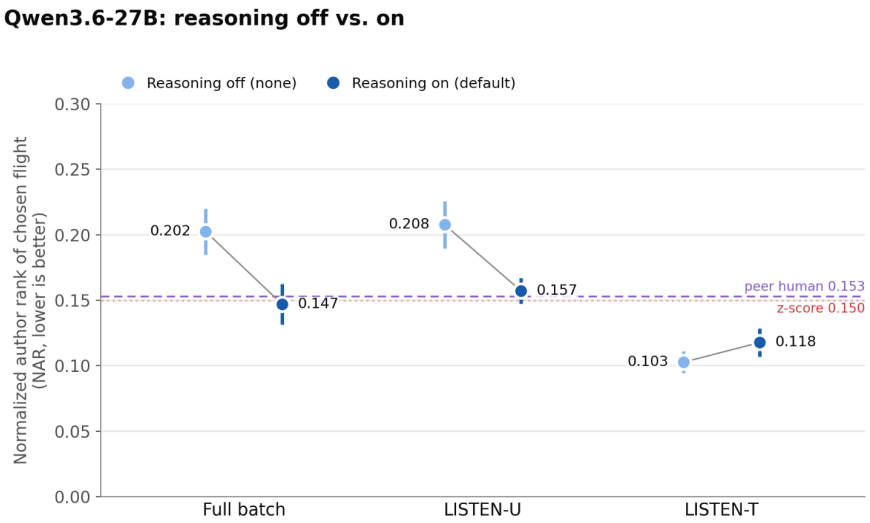
Same full-batch results as the main model comparison. Trend: least-squares fit of log NAR on release date; descriptive only, it does not control for model size or settings.

Reasoning Models create ad-hoc structure

In Open models like Qwen3.6-27B allow us to see the reasoning traces.

We see reasoning significantly improves performance.

In the reasoning trace we see the LLM trying and evaluating different filters and tradeoffs.



IN PROGRESS: reasoning on has 6 of 10 seeds so far.
10 human flight prompts × harness seeds 1234–1243, same candidate order at both levels; dot = mean, bar = ±1 SE across seeds.
Failed calls keep average-unranked NAR.

Inside one full-batch reasoning trace:

85 flights
Full set of options

↓

41 flights
PHX airport

↓

27 flights
Afternoon / evening

↓

2 flights
Nonstop

The model compares the finalists and selects the writer's **#1 flight.**

One qualitative example of internal filtering. A one-call interface can contain many reasoning steps; it need not implement LISTEN-U.

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Ongoing work: broader preference data

Do the original findings generalize across people, tasks, and models?

Collect more data on different tasks

More complex selection tasks

Tokenomics

Most cost and time efficient ways to get performance

Simulated preference data

Using LLMs to simulate different users

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Thanks for LISTENing

Special thanks to Professor Frazier and Professor Shmoys

Original paper

Check out our website